

Plush Toy Factory SCADA Platform

Real-Time Industrial Telemetry & Analytics (7×7 Technical Briefing)

Engineering Presentation

Software Engineering Division

10-Minute Technical Walkthrough

1. Problem Statement: Industrial Blind Spots

- Manufacturing faces continuous product scrap risk.
- Undetected equipment overheating ruins product batches.
- Traditional manual paper logs cause delays.
- Commercial SCADA software costs thousands annually.
- Real-time digital telemetry is mission-critical.

2. The Solution: Zero-License Architecture

- Lightweight open-source industrial telemetry stack.
- Python-driven edge ingestion and simulation engine.
- Embedded SQLite database for rapid persistence.
- Streamlit and Altair interactive browser dashboards.
- Multi-channel reporting via CSV and PNG.

3. Physical Factory vs. Software Models

- Station 1: Sewing line makes Teddy Bears.
- Station 2: Stuffing line fills Husky Dogs.
- Station 3: Embroidery stitches Giant Pandas.
- Simulated PLCs track hourly operational pulses.
- Normalized relational tables map machine assets.

4. Relational Data Architecture (SQLite)

- Normalized schema in Third Normal Form.
- Table `lineas` stores machine asset metadata.
- Table `produccion_scada` stores time-series telemetry.
- Foreign key constraints enforce relational integrity.
- Parameterized queries eliminate SQL injection vulnerabilities.

5. Key Industrial Metrics (KPIs)

- Total throughput: Good completed units produced.
- Scrap rate: Percentage of defective items.
- Dynamic threshold: Over 5 percent warns.
- Tool temperature: Machine head thermal health.
- Real-time delta indicators guide floor supervisors.

6. Interactive Web Dashboard (Streamlit)

- Dynamic dropdown selects specific factory lines.
- On-demand PLC pulse simulation button.
- Altair dual-series trend line visualization.
- Color-coded machine temperature area chart.
- Raw database inspection table view.

7. Automated Multi-Channel Reporting

- Headless Matplotlib script generates PNG graphs.
- Instant visual artifacts for shift handovers.
- Automated CSV exporter for financial ERPs.
- Embedded Excel formulas calculate defect rates.
- Lightweight HTML5 and Chart.js client.

8. SWE Best Practices & Reliability

- Separation of Concerns across software modules.
- Defensive programming prevents division by zero.
- Idempotent database scripts enable safe resets.
- PEP 8 compliance with type hints.
- Git feature-branch workflow with conventional commits.

9. Conclusion & Live Demonstration

- End-to-end SCADA telemetry operating seamlessly.
- Zero license costs with maximum flexibility.
- Actionable visibility protects production batch quality.
- Architecture easily scales to physical MQTT.
- Ready for interactive live system demonstration.